

EAR FALLS, ON, Canada

WMO: 712950

Lat: 50.631N

Lon: 93.221W

Elev: 1190

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.7	-20.1	-30.6	1.0	-24.4	-26.0	1.3	-19.6	15.6	7.9	14.1	5.2	4.3	270	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	83.1	67.2	80.0	65.1	77.1	63.5	70.4	78.5	68.4	75.5	66.5	73.4	6.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	107.4	74.3	66.0	100.1	71.9	63.9	93.1	70.3	35.2	78.5	33.5	75.9	31.9	73.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.8	11.5	-31.9	88.8	4.9	3.7	-35.4	91.4	-38.3	93.5	-41.0	95.6	-44.5	98.3
			-32.1	73.6	4.7	2.5	-35.5	75.4	-38.3	76.9	-40.9	78.2	-44.3	80.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	3.2	6.2	21.0	35.8	48.6	59.8	65.6	63.3	54.2	40.4	24.6	8.1
	DBStd	24.23	14.03	12.45	13.13	10.00	8.77	6.52	5.86	6.25	8.42	8.51	11.37	13.15
	HDD50	6583	1450	1228	900	433	133	6	0	1	52	319	761	1300
	HDD65	10738	1915	1648	1363	877	510	179	63	107	336	764	1211	1765
	CDD50	1492	0	0	1	6	91	299	484	413	177	21	0	0
	CDD65	172	0	0	0	0	3	22	82	54	11	0	0	0
	CDH74	1404	0	0	0	3	42	215	681	393	69	1	0	0
	CDH80	276	0	0	0	0	4	29	150	85	8	0	0	0
	WSAvg	5.0	5.1	5.1	5.3	5.3	5.4	4.7	4.4	4.5	4.8	5.4	5.4	4.9
	PrecAvg	28.0	1.3	1.0	1.4	1.6	2.4	3.8	3.7	4.0	3.0	2.2	1.7	1.3
(15) Precipitation	PrecMax	34.0	3.4	2.6	3.3	4.4	4.9	6.8	7.8	7.8	5.7	6.1	4.1	2.5
	PrecMin	19.1	0.2	0.2	0.1	0.0	0.4	1.4	1.6	0.8	0.8	0.4	0.1	0.5
	PrecStd	4.6	0.8	0.6	0.8	1.1	1.1	1.4	1.5	1.9	1.5	1.4	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	37.8	56.6	67.1	79.1	83.9	89.1	87.0	80.9	69.9	54.4	36.9
		MCWB	34.4	33.8	47.0	50.0	58.6	66.8	70.4	69.3	66.2	57.5	48.2	34.6
	2%	DB	32.3	33.4	48.4	60.4	73.6	79.1	83.7	82.2	75.3	62.8	47.8	32.6
		MCWB	31.3	30.8	40.0	46.4	55.1	62.3	67.8	67.6	64.2	53.7	42.9	31.4
	5%	DB	27.8	30.1	42.3	56.1	69.3	76.2	80.8	78.4	71.1	58.0	43.2	30.2
		MCWB	26.5	27.8	36.2	44.0	54.0	61.3	65.8	65.5	61.8	50.5	40.0	29.2
	10%	DB	23.4	24.9	37.8	52.0	64.6	72.7	77.7	74.8	67.4	53.3	39.3	27.2
		MCWB	22.3	22.7	33.3	41.8	52.0	59.5	64.3	63.8	59.6	48.8	36.7	26.4
	0.4%	WB	34.6	34.4	48.6	52.0	63.2	71.5	73.7	73.2	69.3	59.6	49.2	34.8
		MCDB	35.5	37.1	53.0	62.8	71.5	79.3	84.1	80.3	75.0	65.4	53.3	36.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.0	31.5	41.0	48.3	59.4	66.8	70.8	69.9	66.3	54.9	44.1	31.8
		MCDB	32.3	33.3	46.6	58.4	67.8	74.0	79.5	77.7	72.2	60.8	46.7	32.5
	5%	WB	26.7	27.7	36.9	45.1	56.6	64.2	68.9	67.9	63.6	52.0	40.0	29.3
		MCDB	27.7	30.0	41.9	54.1	65.0	70.6	76.4	75.1	69.3	57.0	42.6	30.1
	10%	WB	22.3	22.9	33.8	42.5	53.9	62.0	66.9	65.9	60.7	48.2	36.7	26.4
		MCDB	23.4	24.7	36.7	51.3	62.6	69.0	73.8	72.8	66.1	52.5	39.4	27.2
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	18.2	19.4	19.5	19.8	18.8	18.9	17.7	16.0	12.2	11.2	13.2
		MCDBR	15.0	17.3	20.6	26.0	27.6	24.6	23.3	22.3	21.2	20.6	15.1	12.6
		MCWBR	13.8	14.3	13.4	14.6	13.0	11.0	9.9	10.2	11.4	12.3	10.6	11.5
	5% WB	MCDBR	14.4	15.6	19.4	24.0	21.1	17.8	19.1	19.1	18.2	18.4	13.7	12.1
		MCWBR	13.4	13.2	13.5	14.7	11.5	10.3	9.7	10.2	11.4	12.5	10.1	11.3
(40) Clear-Sky Solar Irradiance	taub		0.254	0.260	0.288	0.328	0.346	0.364	0.388	0.362	0.342	0.295	0.276	0.247
	taud		2.280	2.288	2.332	2.403	2.455	2.434	2.381	2.464	2.509	2.604	2.460	2.332
	Ebn at Noon		250	281	292	291	289	283	274	277	271	269	243	237
	Edh at Noon		21	27	32	34	34	35	36	31	27	20	18	17
(44) All-Sky Solar Radiation	RadAvg		369	756	1155	1598	1686	1719	1776	1485	1060	604	319	254
	RadStd		43	68	99	114	105	141	145	130	110	75	34	31

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page